

9	(a) (by Newton's third law) force on wire is up(wards) by (Fleming's) left-hand rule/right-hand slap rule to give current in direction left to right <u>shown on diagram</u>	M1 A1 A1	[3]
	(b) force $\propto$ current or $F = BIL$ or $B (= 0.080/6.0L) = 1/75L$	C1	
	maximum current $= 2.5 \times \sqrt{2}$ $= 3.54 \text{ A}$	C1	
	maximum force in one direction $= (3.54/6.0) \times 0.080$ $= 0.047 \text{ N}$	C1	
	difference $(= 2 \times 0.047) = 0.094 \text{ N}$ or		
	force varies from 0.047 N upwards to 0.047 N downwards	A1	[4]